Chapter-1

Introduction to Pattern Recognition

2025-11-05

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PR Pattern Recognition

Pattern – repetition

A **pattern** is a regularity or structure that repeats or follows a certain rule.
A pattern is something organized and predictable.
It can be visual, auditory, mathematical, or behavioral.
Examples:

- **Visual:** stripes on a zebra, tiles on a floor.
- **Mathematical:** sequence 2, 4, 6, 8...
- **Language:** grammar rules.
- **Nature:** leaf arrangements, snowflakes.

Pattern recognition is the process of identifying and classifying patterns from data.

Examples:

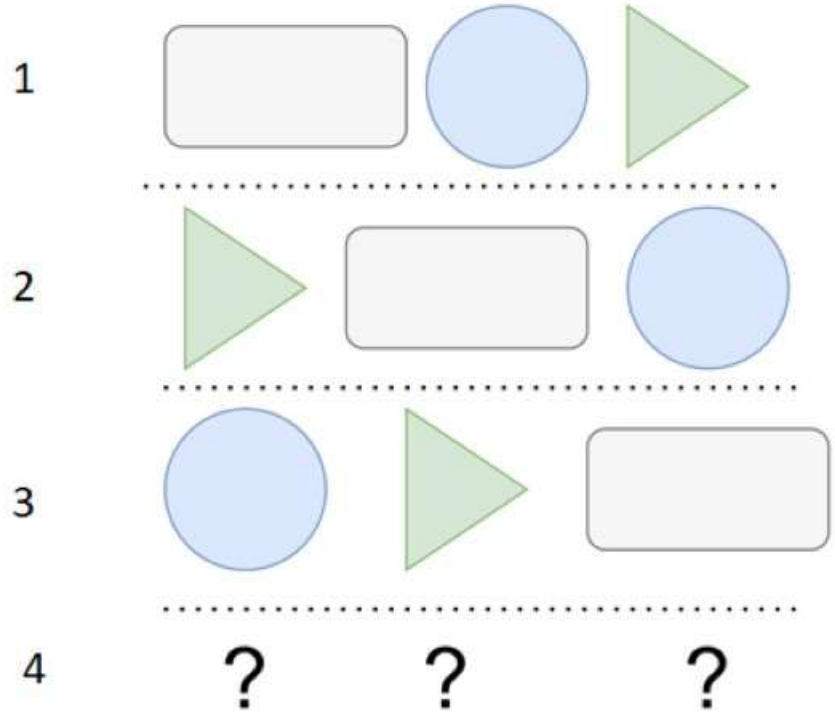
- **Face recognition:** recognizing a person's face.
- **Speech recognition:** understanding spoken words.
- **Handwriting recognition:** digitizing written text.
- **Medical imaging:** detecting tumors in X-rays.

When the monster came, Lola, like the peppered moth and the arctic hare, remained motionless and undetected. Harold, of course, was immediately devoured.

Pattern Recognition

Let's start with an example:

- Can you identify a pattern in this picture and predict the arrangement of the shapes in the row number 4?



Pattern Recognition

You might have already guessed the right answer!

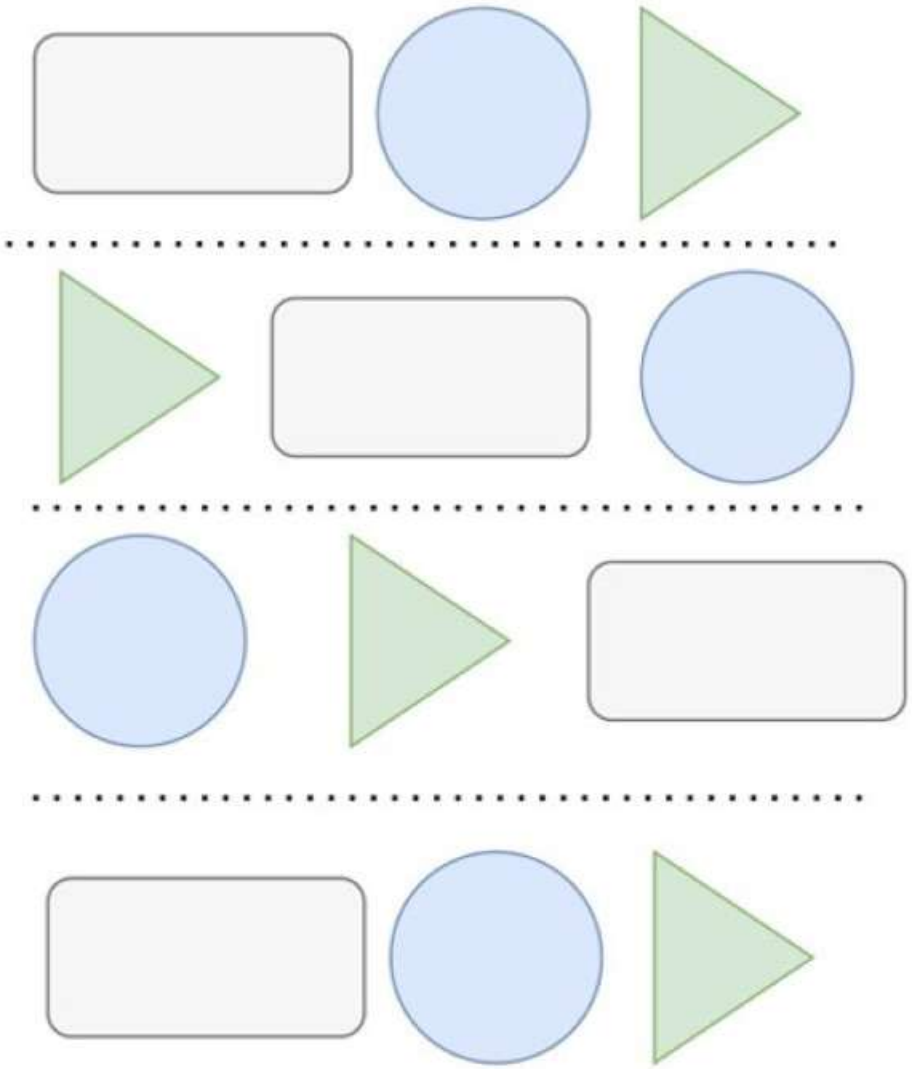
How did we predict the arrangement?

- We observed the **similarities** and **differences** between the first three rows and predicted the arrangement (solved the problems based on a developed pattern).

What are the observed similarities and differences?

Similarities: the unique shapes and colors and direction of the shift (right shift in shapes of every row)

Differences: The position of shapes in each row



What Do You See?

This Yin-Yang Symbol

Figure-ground illusion

PR Pattern Recognition

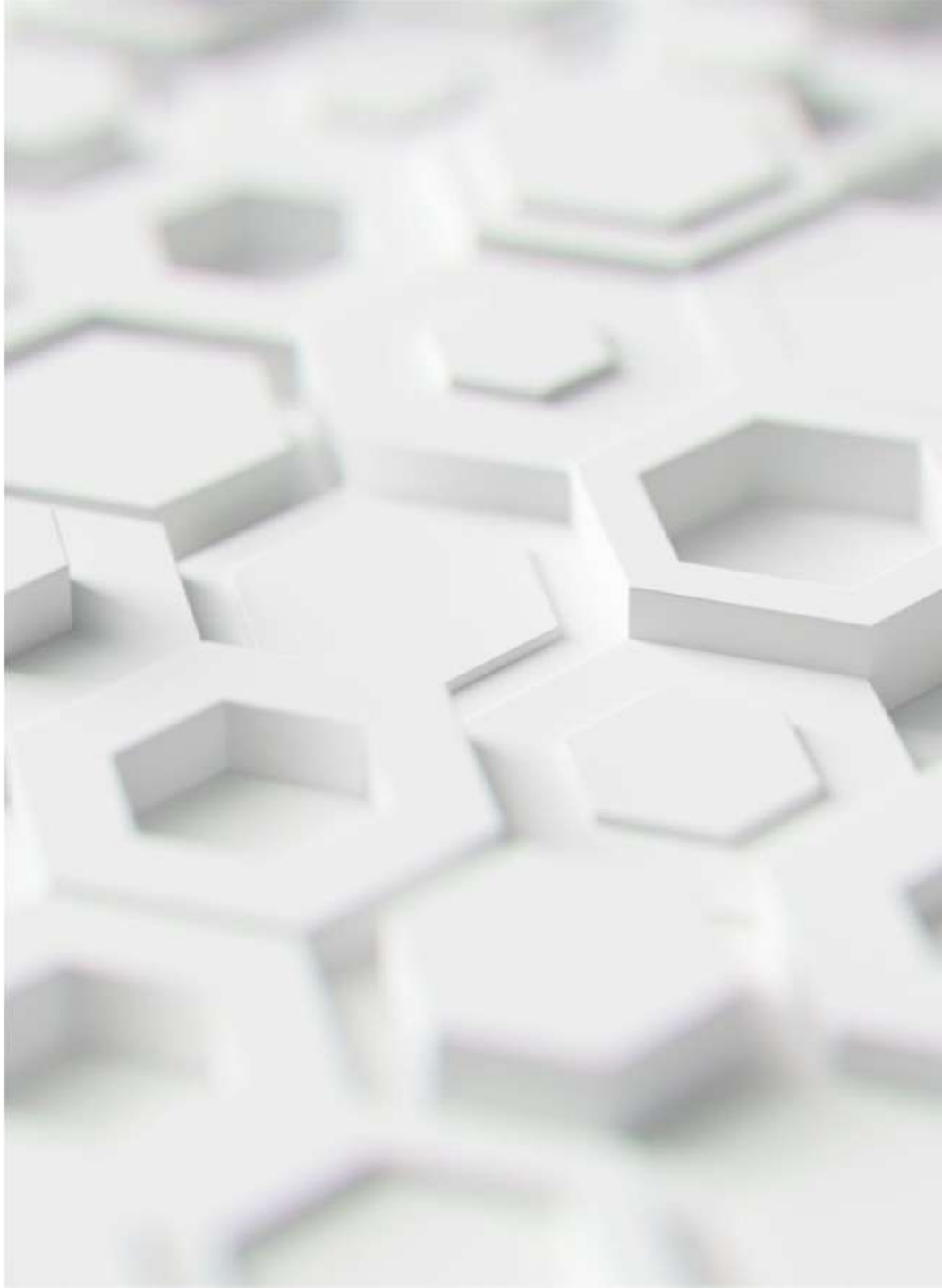
Pattern recognition is a branch of artificial intelligence (AI) and machine learning (ML) that focuses on the **automatic identification and classification of patterns** within data. This data can come in various forms, including:

- **Visual:** Images, videos, medical scans, remote sensing data.
- **Audio:** Speech, music, environmental sounds.
- **Text:** Documents, emails, social media posts.
- **Sensor data:** Temperature, pressure, movement, electrical signals.
- **Time series data:** Stock prices, weather data, medical readings.

Pattern Recognition

- Pattern recognition is the skill of recognizing the similarities and differences between concepts and objects.
- Pattern recognition is the ability to analyze and identify the shared characteristics between parts of a decomposed problem.
- In artificial intelligence, we use pattern recognition to analyze data and identify similarities to recommend an object or content.

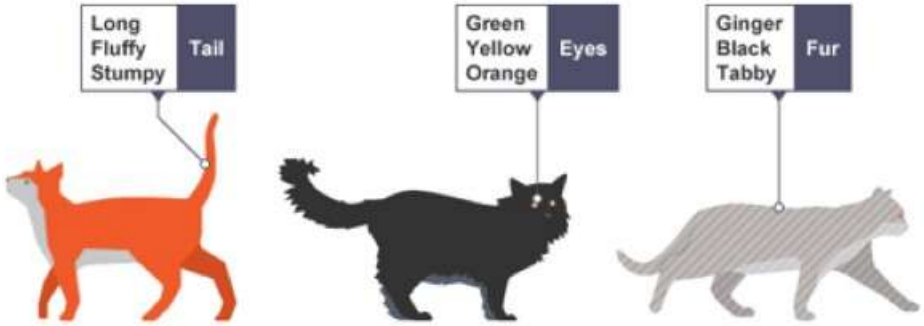
- **For example**, Amazon, Netflix, or Facebook recommend items, movies, and friends or events from the patterns they develop based on your interaction history with these systems.



Pattern recognition

- We often find patterns among the smaller problems we create.
- The patterns are similarities or characteristics that some of the problems share.

Cat
Tail Eyes Fur



Objects have….

Attributes

or….

Properties

Pattern Recognition in Everyday life

Similar to other computational thinking skills, we apply pattern recognition in our daily tasks involuntarily and without being conscious of it.

For example: when you drive to a new city you can easily navigate the streets and follow the traffic signs because you have already created a pattern of traffic signs in your mind and can easily apply/map them in different situations.

When you learn how to make coffee with one coffee machine, its pattern is developed in your mind and you will know how to make coffee with different brands of coffee machines since you replicate the same pattern in different situations.

Pattern recognition major steps

A. Data Acquisition (Sensing)

- Collect raw data from sensors, images, audio, text, etc.
- Example: A camera capturing a handwritten digit.

B. Preprocessing

- Clean and prepare data to remove noise or irrelevant details.
- Techniques: filtering, normalization, background removal, resizing.
- Example: Removing background noise in a speech signal.

C. Feature Extraction

- Extract the most important characteristics (features) that represent the pattern.
- Examples:
  - Edges, corners, textures (in images).
  - Frequency, pitch, duration (in speech).
  - Statistical values, word embeddings (in text).

D. Model training: Develop a model (e.g., neural network, decision tree) that can learn to identify and classify patterns based on the extracted features.

E. Classification and prediction: Once trained, the model can be used to classify new, unseen data points based on the learned patterns.

Applications of PR

Pattern recognition finds application in diverse domains, revolutionizing how we interact with and extract meaning from data.

- **Machine Learning:** The ability of a machine to improve its performance based on previous results.
- **Machine Understanding:** acting on the intentions of the user generating the data.

Here are some of its significant applications:

- Security and surveillance
- Healthcare and medical diagnosis
- Robotics and automation
- Entertainment and media
- Business and finance
- Scientific research and exploration

Applications

1. Security and Surveillance

- **Facial recognition:** Identifying individuals in images or videos for security purposes, access control, and personalized experiences.
- **Anomaly detection:** Detecting suspicious activities, objects, or behavior in real-time using video surveillance footage.
- **Fraud detection:** Identifying fraudulent transactions and activities in financial data, protecting users from financial harm.

2. Healthcare and medical diagnosis

- **Medical image analysis:** Identifying diseases and abnormalities in medical scans like X-rays, MRIs, and CT scans, aiding early diagnosis and treatment planning.
- **Drug discovery:** Analyzing molecular structures and predicting their potential effectiveness in drug development.
- **Personalized medicine:** Analyzing patient data and genetic makeup to tailor treatment plans and improve treatment outcomes.

Applications

3. Robotics and automation

- **Object recognition and picking:** Enabling robots to identify and manipulate objects in their environment, automating tasks in manufacturing, logistics, and warehouses.
- **Self-driving cars:** Recognizing traffic signs, pedestrians, and other vehicles, allowing autonomous driving systems to navigate safely.
- **Robot navigation:** Identifying obstacles and paths in real-time, enabling robots to navigate their surroundings efficiently.

4. Entertainment and media

- **Content recommendation:** Analyzing user data and preferences to suggest personalized recommendations for music, movies, and other forms of entertainment.
- **Image and video editing:** Automatically detecting objects and removing unwanted elements in images and videos, enhancing editing capabilities.
- **Speech recognition and voice assistants:** Converting spoken language into text, enabling natural language interaction with devices and systems.

Applications

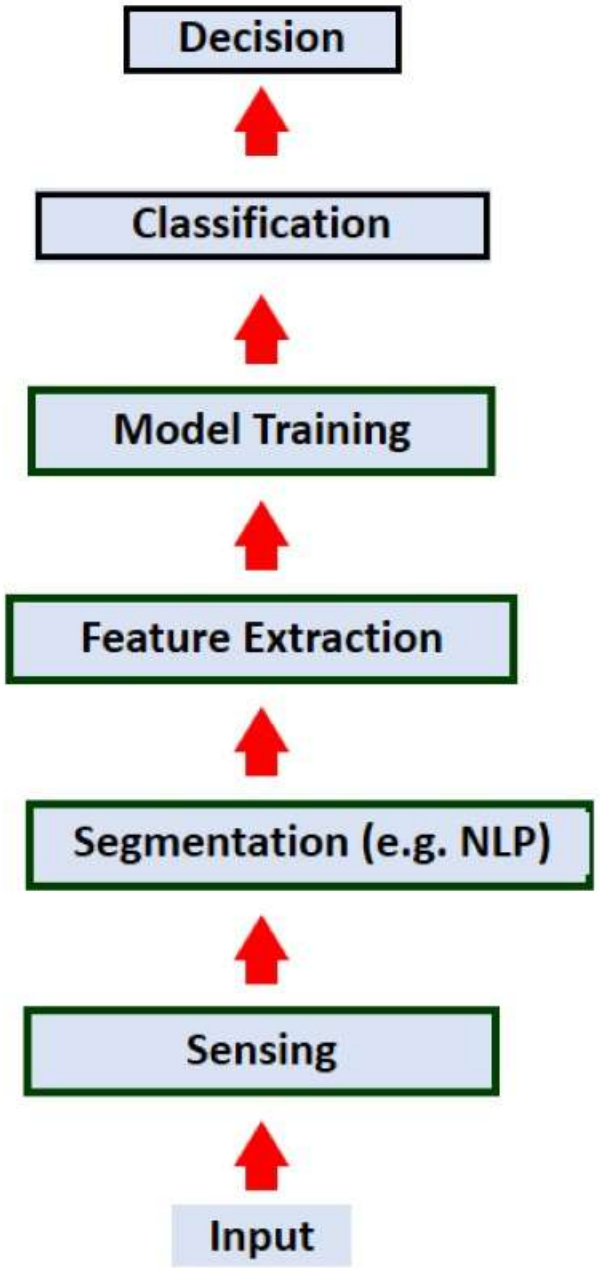
5. Business and finance

- **Market analysis and trend prediction:** Identifying patterns in market data to predict trends, optimize investment strategies, and gain a competitive edge.
- **Customer segmentation and targeting:** Analyzing customer data to group customers into specific segments and personalize marketing campaigns for increased effectiveness.
- **Product design and quality control:** Identifying defects and inconsistencies in products during manufacturing, ensuring quality control and reducing costs.

6. Scientific research and exploration

- **Genomic analysis:** Analyzing DNA sequences to identify genetic variations and their potential role in diseases or other biological processes.
- **Astronomy and astrophysics:** Analyzing astronomical data to identify galaxies, stars, and other celestial objects, furthering our understanding of the universe.
- **Environmental monitoring:** Analyzing patterns in environmental data like satellite imagery and sensor readings to monitor air quality, deforestation, and climate change.

PR Process Flow Decomposition



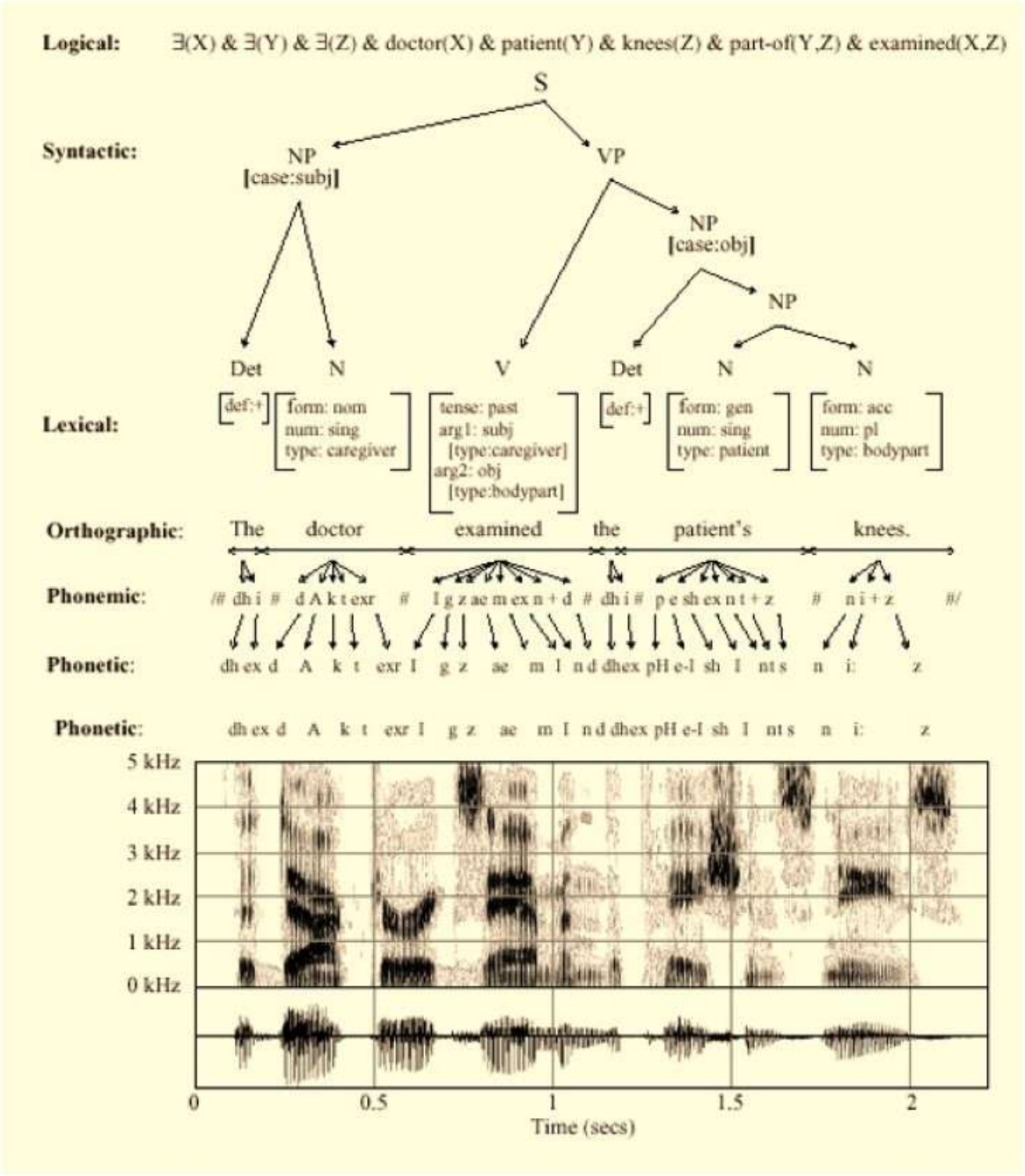
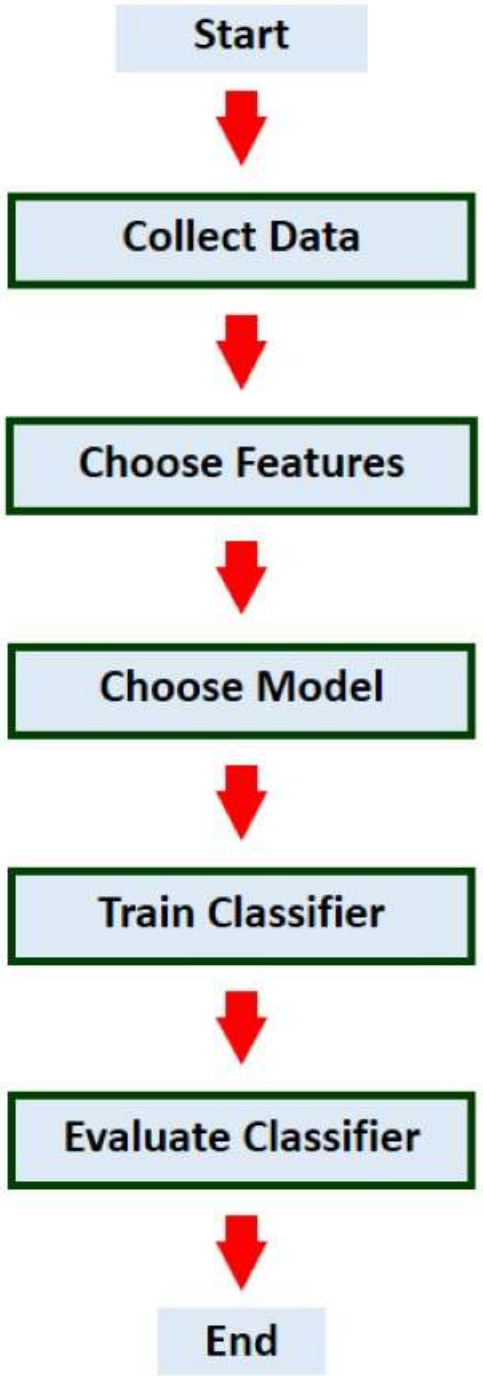


Fig. Speech Audio Process Flow Decomposition

The Design Cycle



Key issues:

- "There is no data like more data."
- Perceptually-meaningful features?
- How do we find the best model?
- How do we estimate parameters?
- How do we evaluate performance?

Goal of the course:

- Introduce you to mathematically rigorous ways to train and evaluate models.

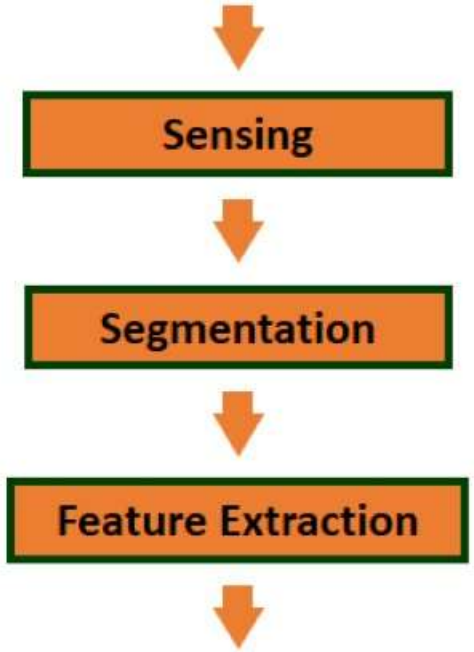
Common Mistakes

- I got 100% accuracy on...
  - Almost any algorithm works some of the time, but few real-world problems have ever been completely solved.
  - Training on the evaluation data is forbidden or repeated on training sample.
  - Once you use evaluation data, you should discard it.
- My algorithm is better because...
  - Statistical significance and experimental design play a big role in determining the validity of a result.
  - There is always some probability a random choice of an algorithm will produce a better result.
- Hence, in this course, we will also learn how to evaluate algorithms.

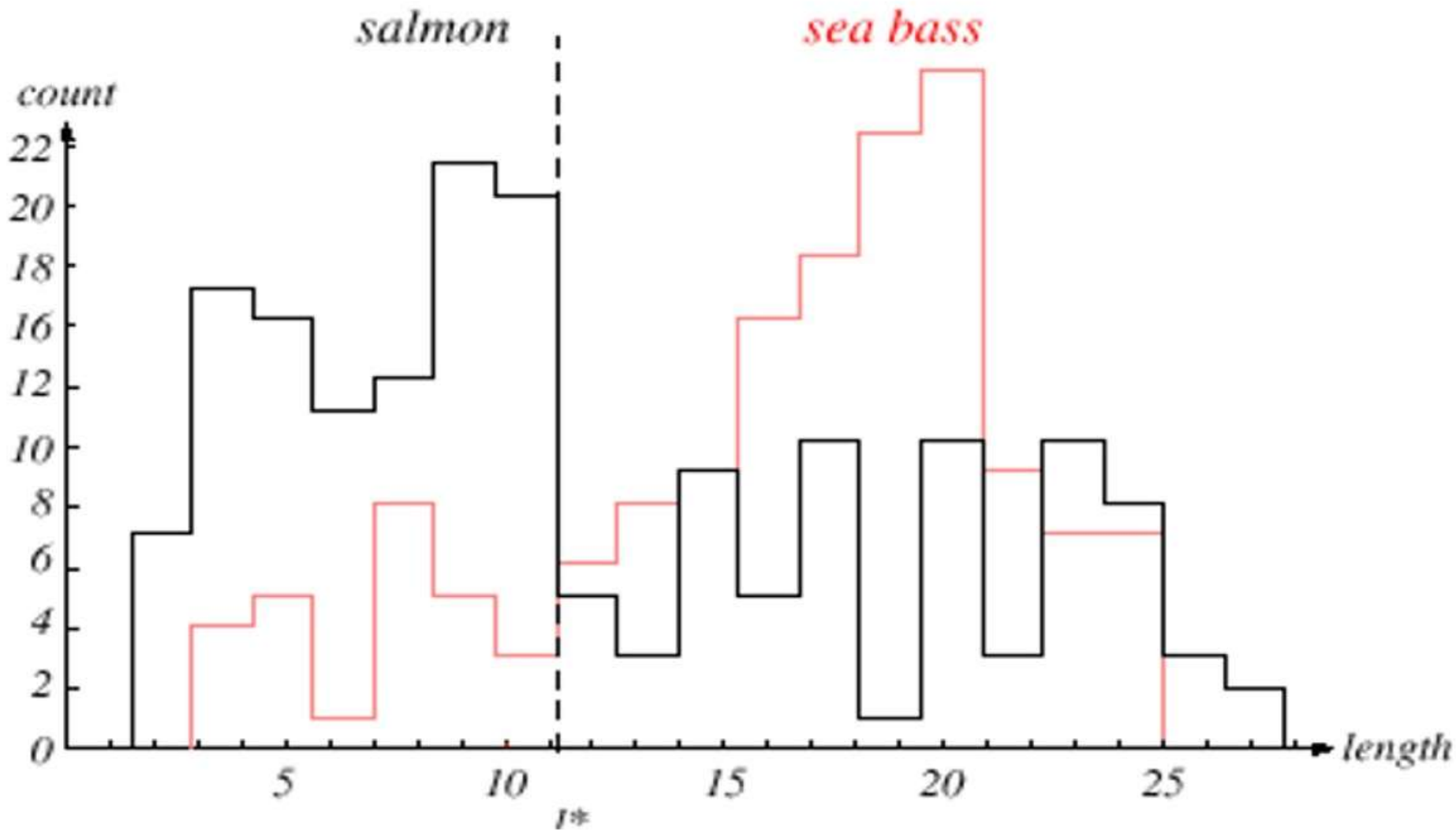
Image Processing Example

- **Sorting Fish:** incoming fish are sorted according to species using optical sensing (sea bass or salmon?)

- **Problem Analysis:**
  - Set up a camera and take some sample images to extract features
  - Consider features such as length, lightness, width, number and shape of fins, position of mouth, etc.

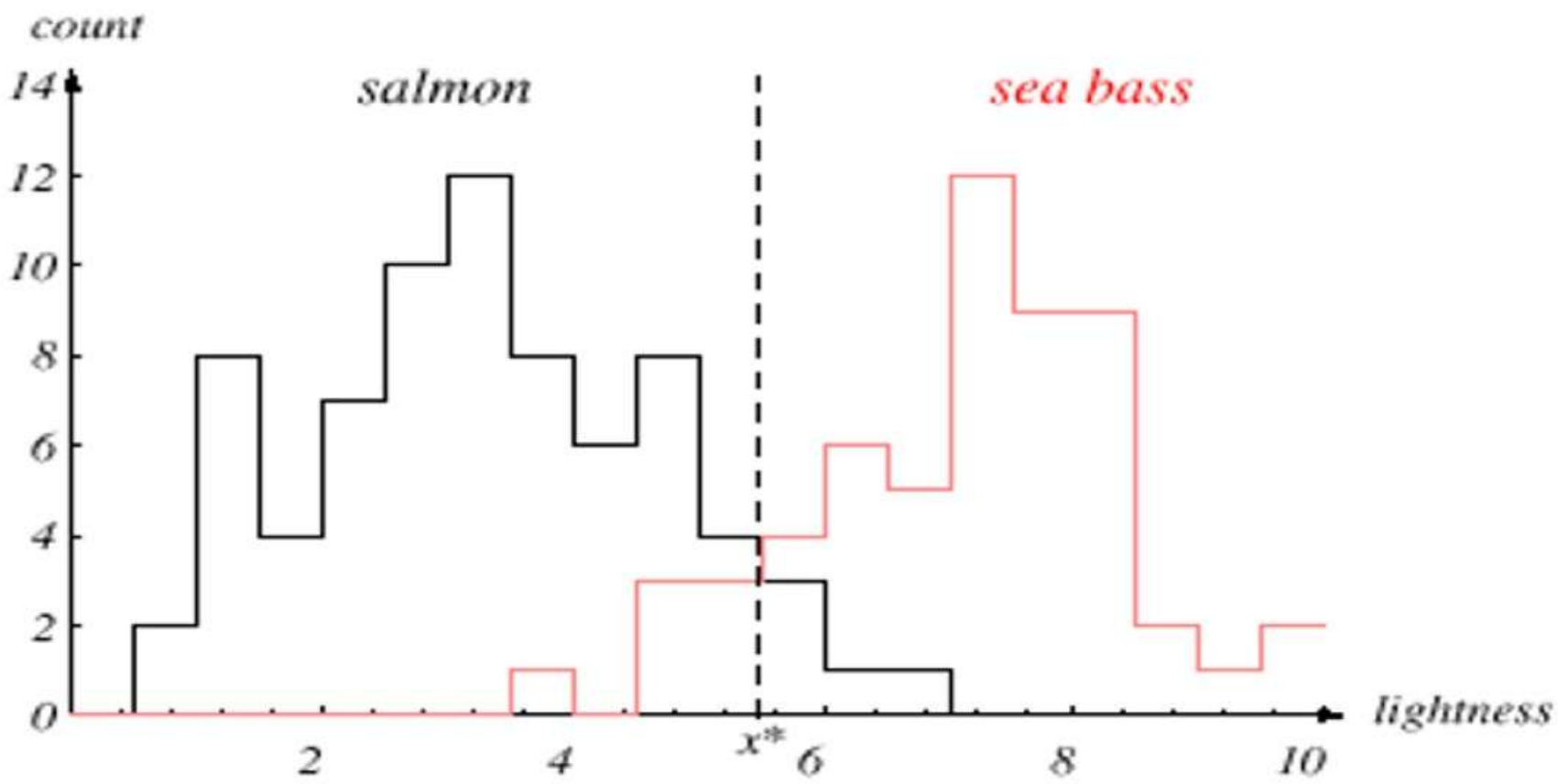


Length As A Discriminator



Conclusion: Length is a poor discriminator

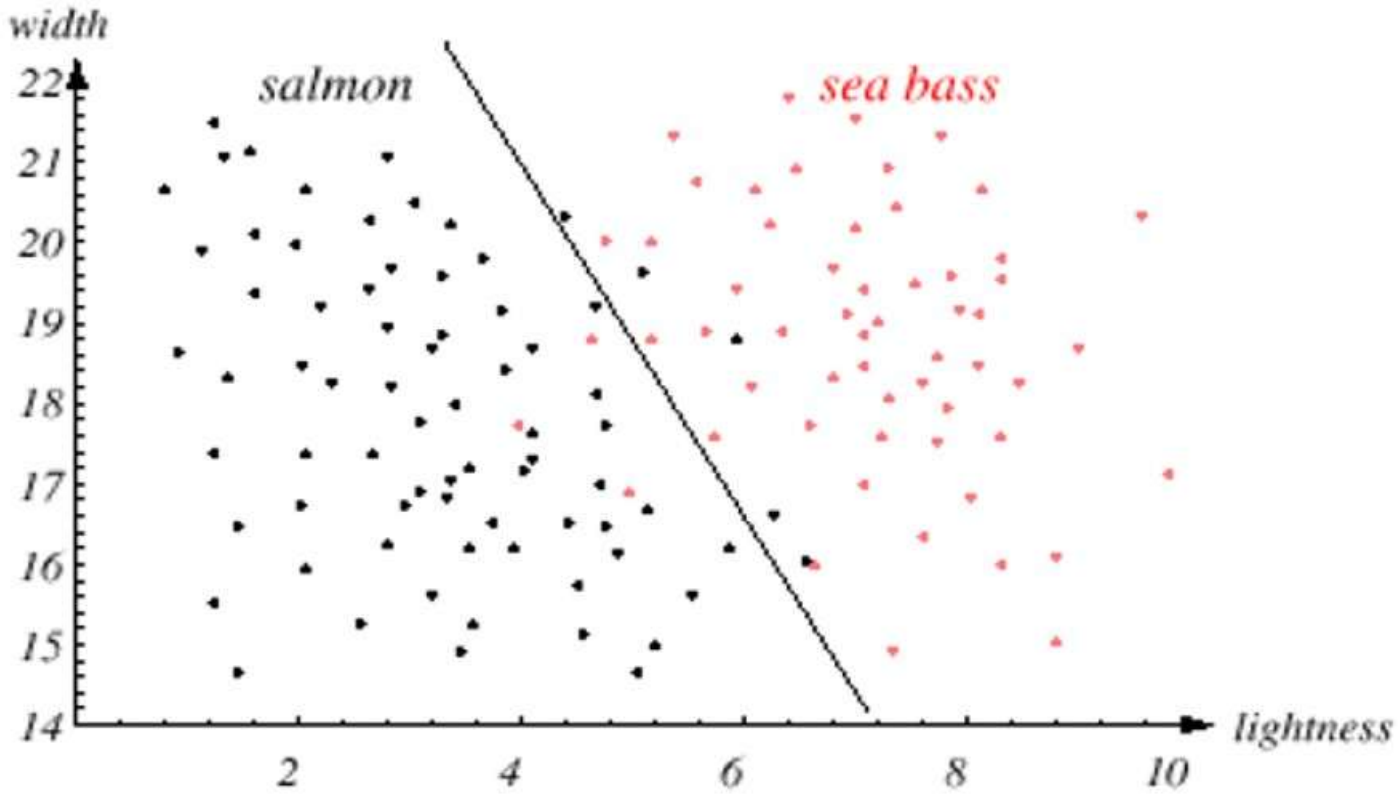
Add Another Feature



- **Lightness** is a better feature than length because it reduces the misclassification error.
- **Can we combine features in such a way that we improve performance?**

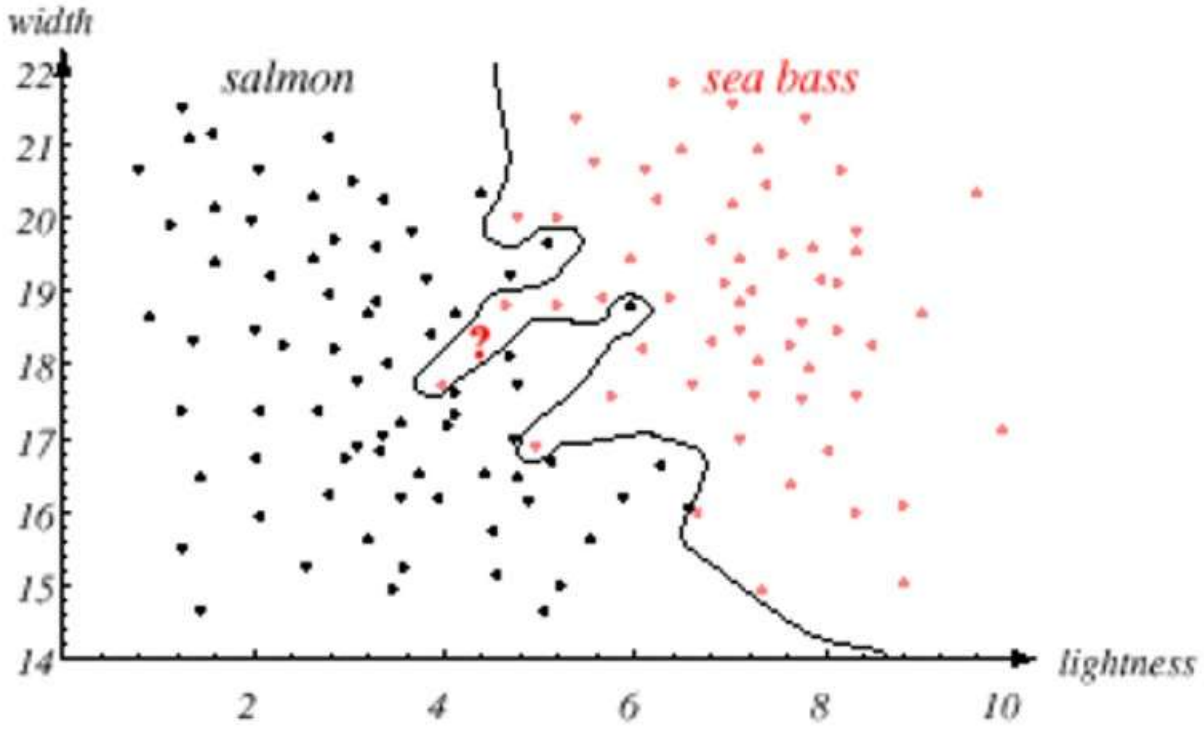
Width And Lightness

- Treat features as a N-tuple (two-dimensional vector)
- Create a scatter plot
- Draw a line (regression) separating the two classes



Decision Theory

- Can we do better than a linear classifier?

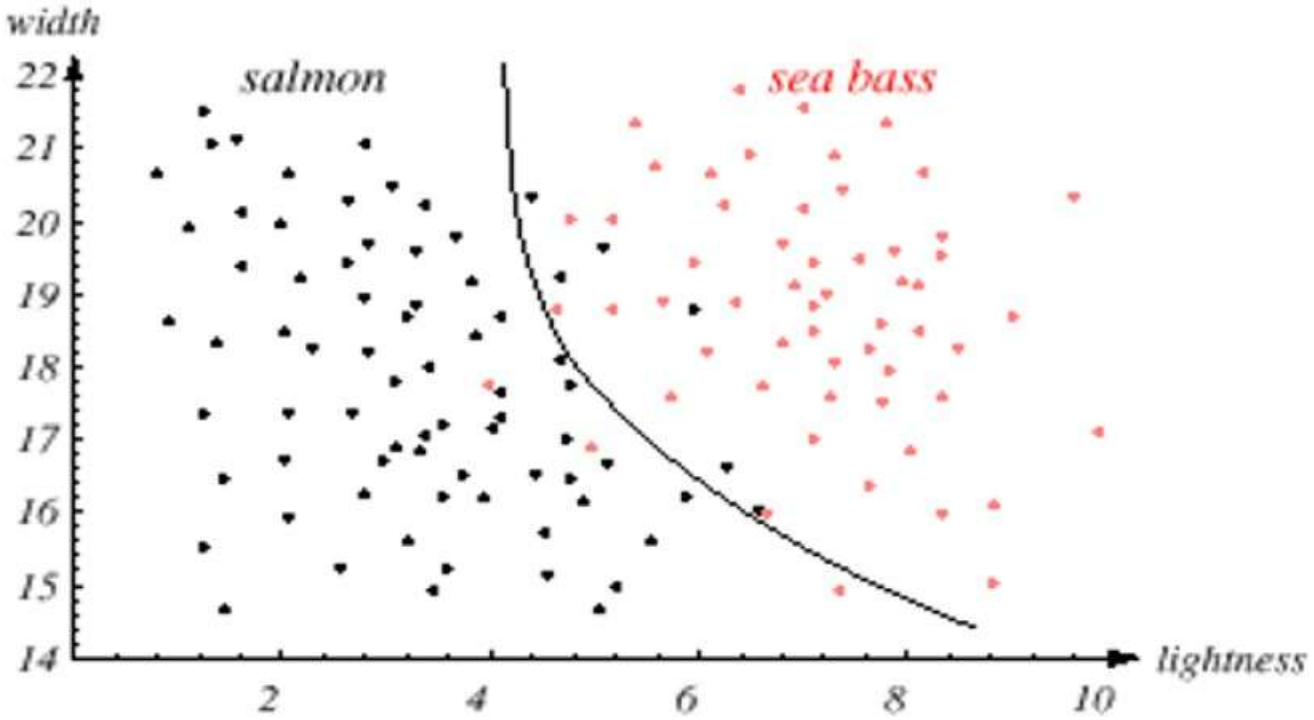


What is wrong with this decision surface?

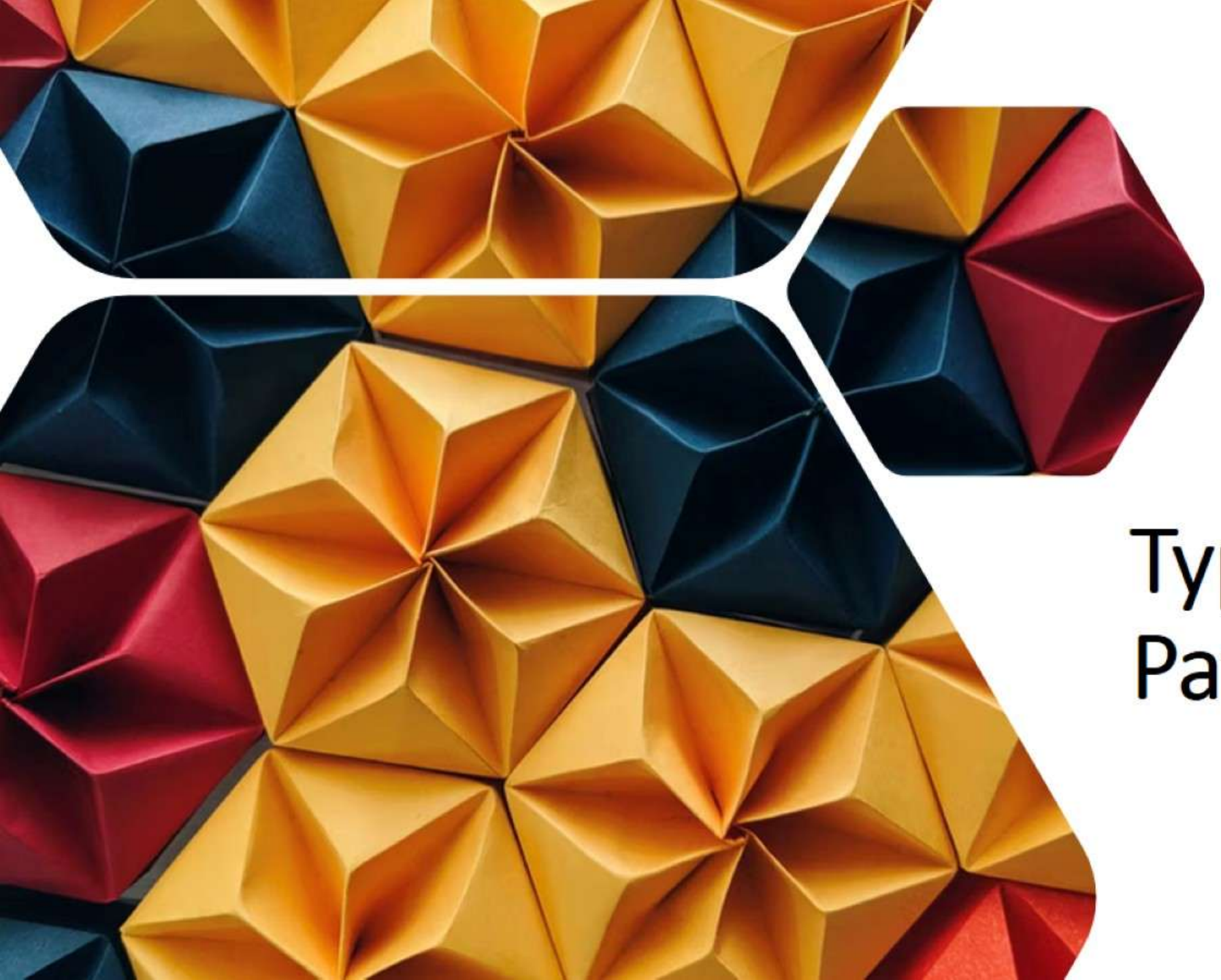
(Hint: generalization)

Generalization and Risk

Why might a smoother decision surface be a better choice?



- This course investigates how to find such "optimal" decision surfaces and how to provide system designers with the tools to make intelligent trade-offs.



Types of Pattern

Patterns

The world around us is filled with patterns, and understanding these patterns is crucial in various fields, from computer science and engineering to biology and music. Here's an overview of different types of patterns:

Type of Patterns	**Definition**	**Examples**	**Applications**
Visual Patterns	Patterns perceived through sight, often involving spatial arrangements of elements like colors, shapes, textures, and objects.	Stripes on a shirt, geometric shapes in a painting, facial features of a person, repetitive patterns in nature (e.g., flower petals, animal markings).	Object recognition, image segmentation, medical diagnosis, autonomous navigation, design and aesthetics.
Acoustic Patterns	Patterns perceived through sound, involving temporal sequences of sound waves with specific characteristics like frequency, amplitude, and rhythm.	Musical melodies, speech rhythms, animal vocalizations, environmental sounds (e.g., ocean waves, traffic noise).	Music genre classification, speech recognition, anomaly detection in machinery, audio fingerprinting, sound effects creation.

Patterns

Patterns often combine elements from different types. For instance, a visual pattern like a traffic light combines specific colors with a temporal pattern of changing states (red, yellow, green).

Type of Patterns	**Definition**	**Examples**	**Applications**
Text Patterns	Patterns in written language, involving sequences of characters, words, and sentences with grammatical and semantic structures.	Grammatical rules, sentence structure, recurring phrases, literary styles, thematic patterns in narratives.	Machine translation, sentiment analysis, plagiarism detection, text summarization, topic modeling.
Sensor Data Patterns	Patterns in data collected by various sensors, including temperature, pressure, movement, and electrical signals.	Periodic fluctuations in temperature readings, recurring patterns in pressure sensor data indicating machine faults, movement patterns captured by motion sensors.	Predictive maintenance, anomaly detection in sensor networks, environmental monitoring, activity recognition, human-computer interaction.
Time Series Patterns	Patterns in data that changes over time, often represented as sequences of values at specific time intervals.	Stock market prices, weather data, sales figures, heart rate readings, electricity consumption.	Time series forecasting, financial analysis, anomaly detection in time series data, medical diagnosis, traffic prediction.

Challenges on high-dimensional visual data

- **Computational Complexity:** High-dimensional data often requires more computational resources for processing and analysis. Algorithms may become computationally intensive, leading to longer processing times and increased memory requirements.
- **Overfitting:** With a large number of features, there is a higher risk of overfitting, where a model captures noise or irrelevant patterns in the data, rather than the underlying true patterns. This can result in poor generalization performance on unseen data.
- **Dimensionality Reduction:** While high-dimensional data provides rich information, it can also lead to redundancy and irrelevant features. Dimensionality reduction techniques are often needed to extract the most informative features and reduce the computational burden.
- **Feature Selection and Extraction:** Identifying relevant features from high-dimensional visual data is crucial for effective pattern recognition. However, feature selection and extraction methods must be carefully chosen to avoid loss of important information.
- **Data Quality and Preprocessing:** High-dimensional visual data may suffer from noise, missing values, and other quality issues. Proper preprocessing steps are essential to clean the data and improve the robustness of pattern recognition algorithms.
- **Visualization:** Visualizing high-dimensional data poses challenges, as human perception is limited to three dimensions. Techniques such as dimensionality reduction and projection methods are required to visualize and interpret the data effectively.
- **Curse of Dimensionality:** As the dimensionality of the visual data increases, the volume of the feature space grows exponentially. This can lead to sparsity of data, making it difficult to effectively capture patterns and relationships.

Thank you!

Let's Connect

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